Section 1: Metric Selection & Technical Framework

1.1 Interpretation of Components

- **True Positives (TP):** Sick patients (Malignant) correctly identified as sick.
- **False Positives (FP):** Healthy patients (Benign) incorrectly flagged as sick.
- **False Negatives (FN):** Sick patients (Malignant) incorrectly told they are healthy. This is the most dangerous error in a medical diagnostic framework.

1.2 Why Accuracy Alone Is Insufficient

Accuracy simply computes \frac{\text{Correct Predictions}}{\text{Total Predictions}}. If a medical dataset consists of 95% benign samples, a naive model that predicts "benign" for every single patient achieves 95% accuracy while completely missing 100% of the active cancer cases. Therefore, tracking targeted metrics like Precision, Recall, and F1-Score is critical.

1.3 Metric Selection for Medical Diagnosis

In medical contexts, **Recall (Sensitivity)** is prioritized.

Recall = TP/TP+EN

A low recall implies high False Negatives, meaning patients with malignant tumors would go undiagnosed. While a low Precision creates anxiety via False Positives, those can be filtered out during secondary clinical checks.

If data shows severe imbalances, F1-Score (harmonic mean of Precision and Recall) serves as a vital comprehensive indicator to balance trade-offs.

Section 2: Imbalance Handling Analysis

Real-world datasets rarely present a clean 50/50 balance. In this task, we addressed potential imbalances using the Class Weight Adjustment method within our Logistic Regression formulation.

By passing class_weight='balanced', the optimization cost function shifts penalization dynamics. It increases the penalty for misclassifying the minority class proportionally to its underrepresentation. This systematically forces the decision boundary to protect the minority class, typically increasing Recall for the critical class at a marginal expense to overall Precision.

Section 3: Final Model Decision & Comparison

Based on the empirical pipeline runs, we contrast our structural setups:

Model Architecture Structural Integrity / Complexity Stability

vs.

Interpretability Overfitting Vulnerability:-

Logistic Regression Linear framework; robust baseline with low parameters.

Extremely high interpretability; coefficients yield clear odds-ratios.

Low vulnerability; regularized easily via penalty settings.

Decision Tree Hierarchical splits; creates high-variance non-linear rules.

Visually easy to read, but highly unstable to minor training sample variations.

High vulnerability; easily grows over-complex without pruning.

Final Decision :-

Logistic Regression (with Balanced Class Weights) is chosen for production delivery. It provides smoother generalization, clear probabilistic confidence bounds via the sigmoid function, and an higher AUC score compared to basic Decision Trees, which struggle with sharp, unpruned regional overfits on tabular dimensional feature spaces.